CHAPTER 9

Blockchain-based e-voting protocols

Srijanee Mookherji, Odelu Vanga and Rajendra Prasath[a]
Indian Institute of Information Technology, Sri City, Andhra Pradesh, India

1. Introduction

Voting is an act of decision-making that plays an important role in choosing either a leader for a designated position or privileging a contending process or an event to take a lead in a specific situation. It provides a platform to express majority by means of choice in scenarios ranging from small-scale boardroom decisions to large-scale democratic decisions. The concept of elections is said to have been introduced in the Ancient Greek and Roman periods. The conventional method of voting can be classified into five broad classes (Lambrinoudakis et al., 2003):

- hand-counted ballot papers and boxes system
- direct-recording mechanical (lever) voting machines
- punch cards as ballots
- optical mark-sense ballots
- direct-recording electronic voting machines

Talking about the hand-counted ballot paper and boxes system, the Australian Ballot system was embraced by almost every government around the world. Its uniqueness was in the fact that ballots were structured and issued by the government and voters voted in secret (Miller, 1995; Fredman, 1967; Brent, 2006). But, it required innumerous papers, infrastructure, and manpower. In addition, it needed a significant amount of time to count the ballots, and human errors would cause discrepancies in the result. Along with these, many security flaws like lack of voter anonymity, vote integrity, etc., were present. Unethical activities, such as vote manipulation, miscalculation of ballots, biased result declaration, and voter impersonation, took place. This resulted in questioning the accuracy and integrity of such a voting system.

[a] This work was supported by the seed grant funded by IIIT Sri City, Chittoor.

Blockchain Technology for Emerging Applications
ISBN 978-0-323-90193-2
https://doi.org/10.1016/B978-0-323-90193-2.00006-5

To overcome the count bias and delay in result publishing, mechanical lever voting machines were introduced and brought into practice in the 1930s in the United States. These machines counted the votes the moment a voter cast his choice, thus providing the final results minutes after the polls closed. However, the major issue with these machines was that recounting was an impossible task, as no backup of votes was available. There was just an odometer for rechecking, which kept the total vote count for each candidate (Cranor, 2003; Greene et al., 2006).

In 1964, IBM's Portapunch mechanism was used to design Votomatic that used punch cards as ballot papers. The major drawback of the system was that even though recounting was possible, there was no guarantee that the punches would be clean. Also, there was no intuitive system to decipher the vote cast from a mispunched card. After the controversy surrounding the US presidential election in the State of Florida in 2000, in which punch cards were used, governments around the world began scrutinizing and questioning the integrity of the followed voting systems (Saltman et al., 2003; Kohno et al., 2004; Roth, 1998). This has resulted in the rise of the concept of electronic voting.

The optical mark-sense scanner was introduced in 1950s for college grading purposes, and it was applied in the field of electoral systems in 1970s. The voters were required to color the slots next to their desired candidates using a pen or a pencil on a ballot. The scanner would scan the colored area using visible light and note the choices. This system of voting faced a similar problem of human error where a voter does not mark the ballot properly. In the case of the year 2000 US presidential election, many ballots were challenged due to inappropriate marking of the ballots (Awad and Leiss, 2016).

Electronic voting is the process of using computer devices to help in casting and counting votes. Any voting mechanism that uses electronic devices can be categorized into e-voting techniques. Direct-recording electronic voting machines and internet voting are considered the most widely accepted e-voting techniques (Lauer, 2004). Shoup and Microvote had already come up with direct-recording electronic voting systems in 1986. The advantage of the system was it had a better interface, making it easier for people to vote. Everything being done electronically meant that counting was automated and results could be published faster. The only drawback of the system was submission of empty ballots due to improper pressing of buttons on the machine, resulting in wastage of votes. Since the vote was also electronically transmitted, there was no physical existence of the vote, making it impossible to go for a recount. If a machine was tampered with by malicious parties, then the authorities had no option to know whether the votes were authentic (Miller, 1995; Lauer, 2004; Epstein, 2007).

Internet voting is a technique where a voter can vote securely over the internet from any part of the world. In 1997, David Wolf, an American astronaut, became the first person to cast his vote over the internet. The vote was cast via email from Mir Space Station (Lauer, 2004). With the beginning of internet voting, one can now vote in the comfort of their home, and an increase in voter turnaround can be expected as a result. Also the concept of early voting has come into play, where a voter can vote before the election day at satellite polling stations. The only thing that needs to be maintained is the fact that the ballots should be kept uncounted and sealed till the end of the election. With a higher voter turnaround, the frauds that had been occurring due to absentee voting are also expected to decrease (Cranor, 2003; Jones, 2003).

However along with advantages, there are gaps in e-voting that need to be addressed. With most of the e-voting systems being centralized in nature, there is one central trusted authority, and there is a high possibility of a rigged administrator turning the entire voting system into a biased one. This also results in integrity issues and a lack of trust in the voting system from the voter's side. A solution to such a problem can be solved by using a decentralized data management system. The concept of blockchain can provide something similar to the requirement. Blockchain is a decentralized, transparent, and distributed ledger that is immutable in nature.

Blockchain-based e-voting protocol can be implemented in many ways. The main idea is to maintain privacy of votes and voters and integrity of the election. There are currently many researches going on to incorporate blockchain in e-voting. Most of the proposed works are based on permissionless blockchain (Bistarelli et al., 2017), but there are a few issues that arise that are discussed further in the chapter. Again, works (Hjalmarsson et al., 2018; Lee et al., 2016) have used permissioned blockchain. And, there are protocols (McCorry et al., 2017; Liu and Wang, 2017) that function without a trusted third party (TTP). Simultaneously, there are protocols (Lee et al., 2016) that have used the help of TTPs. There are different techniques used by researchers to ensure anonymity and keep votes and information a secret. Others (Hjalmarsson et al., 2018; McCorry et al., 2017) have used the concept of zero-knowledge proof for information hiding purpose. Wang et al. (2018) used ring signature to ensure anonymity. Many different techniques are being combined by researchers around to world to propose a secure blockchain-based protocol. A few of these are summarized in the upcoming sections, and a comparative study for the works is provided.

This book chapter is organized as follows: Section 2 describes the onset of electronic voting and how it evolved over the time. We further discuss the various security properties and performance properties required for a secure electronic voting system. A comparative study of existing research is made based on the aforementioned properties. Finally, issues in electronic voting systems are discussed. Section 3 describes the advances in blockchain-based e-voting systems by going through the significant contributions toward developing such systems. A comparative study of various works is presented, and a discussion on the prevalent issues of blockchain-based systems is provided. Section 4 concludes the book chapter.

2. Electronic voting

An election is an indispensable part of an efficiently functioning modern democracy. An election is a medium to assess the trust of people in their government. As mentioned, after the disruption in the 2000 US presidential election, citizens began questioning different aspects of the electoral system, which gave rise to thoughts about an alternative for conventional voting. Hence, electronic voting was encouraged, as it would ensure higher voter turnout, making the democratic process stronger and more reliable (Lambrinoudakis et al., 2003).

E-voting is cost-effective and easy to implement. The ease of access to voting increases the voter's interest to take part in democratic decision-making. It assures increased voter participation (Raikov, 2018), and it helps in providing accurate decisions. An e-voting system is expected to the overcome the backdrops of a traditional method and provide a secure way of holding an election with the help of devices and infrastructure that use cryptography techniques to secure the voting process.

Fig. 9.1 depicts various phases and functionalities of the actors present in a general e-voting system.

Let us now look at the general structure of an e-voting system. We begin with the actors who are essential for an e-voting system (Baiardi et al.; Joaquim et al., 2003):

- **Voter**: Voters are responsible for casting a vote for their desired candidate.

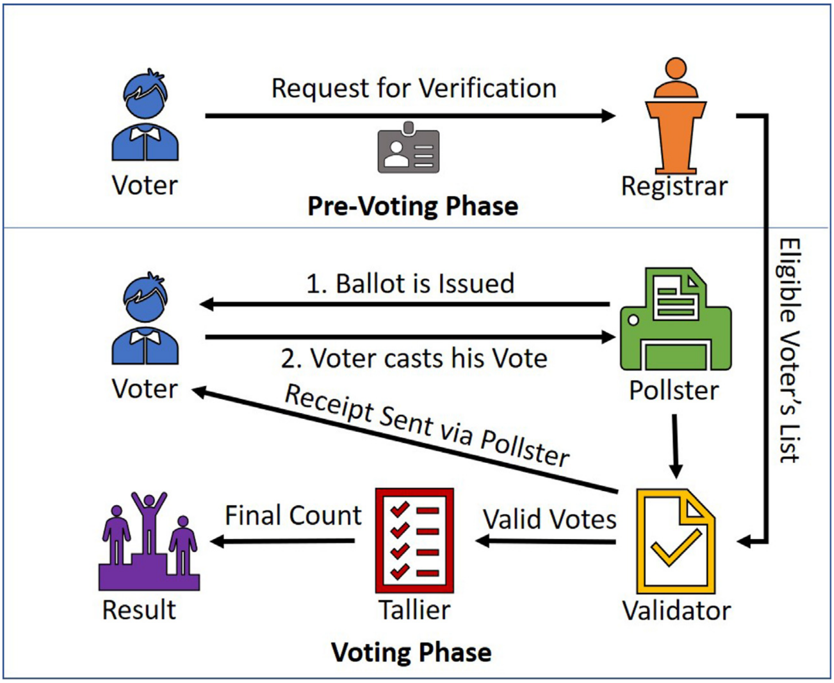

Figure 9.1 Typical phases in an e-voting protocol (Baiardi et al.; Joaquim et al., 2003).

- **Registrar**: The registrar is responsible for the initial setup by registering the voters against a request. In a majority of the systems, only an eligible voter is allowed to register. The registrar holds a list of eligible voters and registered voters. They finally publish a list of verified registered voters, which is used throughout the voting process. The registrar is also responsible for issuing a public key and private key pair to the voters, where henceforth a voter is identified by their identification code and the public key. In many systems, the valid voter list also contains a parameter that is handled by the validator to validate a voter's ballot.
- **Pollster**: After a voter has voted, a pollster is responsible for encrypting the ballot, sending the ballot to the validator, and finally submitting the vote to the ballot box. The voter is required to have trust only in the pollster. The pollster also functions during the time of vote verification by the voter.
- **Validator**: The validator ensures that a voter is valid and has cast their vote only once. Its function is to stop multiple instances of voting.

The validator signs a blind ballot using a validation certificate. The validator uses the valid voters list to check the voter's signature on the ballot using the public key. Once a vote is validated, the parameters on the voter's list are updated. The sequence of vote validation is not recorded. The voter finally unbinds his vote and sends it to the tallier for counting.
- **Tallier**: The tallier verifies the ballot submitted by the voter by checking its uniqueness compared with the other ballots received from other voters. Once the ballot is checked, a receipt is provided to the voter. The tallier is also responsible for counting the votes and publishing the final result.

A voting system is considered secure if it satisfies the following security and performance-related properties:

2.1 Security properties

- **Eligibility**: Only an authorized voter should be allowed to vote. Every voting system should have a fool-proof authentication mechanism.
- **Privacy**: Only a voter should have the knowledge of his choice throughout the voting phase. This property states that the system can keep the voter's choice as a secret even from the concerned authorities.
- **Voter anonymity**: A voter should remain anonymous during the entire time of the election process. It goes hand in hand with privacy, as a vote cast is anonymous and private.
- **Lack of reusability**: A voter can vote only once, and this is used to stop double voting.
- **Verifiability**: This property can be bifurcated into *individual verifiability* and *universal verifiability*. Individual verifiability allows a voter to check if their vote has been counted or not. Universal verifiability ensures that every voter can verify the final result of election at any time after the result has been announced. A more rigid term is end-to-end (E2E) verifiable, which provides a receipt to the voter to ensure that their vote has been noted.
- **Fairness**: This property ensures that the result will be computed only after the end of an election.
- **Lack of coercibility**: This property ensures that vote buying and selling is not possible throughout the process of an election.
- **Receipt-less**: This property entails lack of coercibility, where the voter does not receive any receipt to proving their vote.

2.2 Performance properties

- **Correctness (completeness and soundness)**: Votes should be counted correctly. Correctness can be bifurcated into (1) completeness, which means that all valid votes have to be counted, and (2) soundness, which means that all invalid votes should be excluded from counting.
- **Robustness**: The voting system should function appropriately even when there is a partial failure. It should not be affected by misbehaving voters either.
- **Efficiency**: This is the property that is used to measure the computational complexity of the entire system.
- **Mobility**: This property allows users to vote from mobile devices.

2.3 Significant contributions in e-voting

In 2005, Estonia was the first country to declare that their nationwide election held using a remote internet voting process was successful (Clarke and Martens, 2016). National ID cards issued by the government were used for voter identification. The voters were allowed to vote early over the internet and a chance to vote was open until the election day. To attain the trust of the voters in the system, traditional voting was given a priority, where if a voter chose to go to the polling booth to cast their vote even after casting an e-vote, the e-vote would be deleted and the physical vote would be calculated. The voters were also given an option to re-vote and the old vote would be deleted. If a voter tried to vote twice (which is illegal), only the last vote was considered. An envelope method (Maaten, 2004) was used in the voting system. The voter used their ID card to get into the homepage of the voting website, hosted by the election commission. The ID number was verified against a list of valid IDs. The voter's choice was encrypted, which was the inner envelope. The voter's choice was then confirmed with a digital signature, making it the outer envelope. The outer envelope was removed during counting, and the final tally was done anonymously by the election commission.

The first e-voting scheme as a concept was proposed by Chaum (1981). In the proposed scheme, the votes are transmitted through anonymous channels, utilizing the concept of mixnets (Chaum, 1981). It does not require a centrally trusted authority, and if one authority remains honest throughout the process of mixes, the system remains secured. Voter privacy is ensured using pseudonyms, and a public-key cryptosystem is used. In this scheme, it is assumed that only registered voters are taking part in the election. The details of each voter are masked using the pseudonym. The pseudonym, in turn, decrypts the vote of a voter. In this proposed system, a mix function is introduced to ensure that a voter can vote only once, so in a single mix, voters submit their ballots. For multiple mixes, a group of ballots is generated that is then processed as a single batch. The ballots are finally counted after checking the pseudonyms. This scheme ensures that no voter can vote using two different pseudonyms. The scheme ensures the privacy of voters and vote verifiability.

Cohen and Fischer (1985) proposed an e-voting protocol that uses homomorphic encryption. A time-stamped public bulletin is used for communication between the voters and electoral committee. Beacons are used to generate pseudo-random bits and create one-way location. The issue with the protocol is that the election authorities can read any vote at any time, thus causing privacy and integrity problems. An extension to that work (Cohen and Fischer, 1985) was done by Cohen (1986) where the government is replaced by multiple tellers (mix servers). This distributes the trust to every teller, and if one of the tellers is compromised, then the voting system would be compromised. Also, this paper gave allowance to the government to publish the winner without actually releasing the tally.

In 1994, Benaloh and Tuinstra (1994) proposed a receipt-free election protocol that provided secret verifiable ballots. The voters could not prove the contents of the ballot. This in turn prevented vote coercion. Cramer et al. (1996) showed a multiauthority voting system. A bulletin board is provided for voters to cast, encrypt, and share their votes. Zero-knowledge proofs, being expensive, were eliminated, and instead, a noninteractive proof of validity was proposed, reducing the computation cost from quadratic to linear. Each vote comes with this proof. A threshold system is used to disable a single authority from decrypting the votes. The proposed system is compatible to binary voting. If a different kind of voting is to be implemented, proof numbers will increase.

In Cramer et al. (1997), a multiauthority voting system is proposed that is based on homomorphic characteristics of the ElGamal (1985) crypto-system and proofs of knowledge. The decryption private key is shared to all authorities using a threshold system. Authorities here commit to publicly share their secrets to avoid malicious activities. For decryption, more than half of the authorities are required. In case of binary elections, linear cost is taken with respect to the number of voters. However, a multiway election is computationally a costly affair.

In Juang et al. (2002), a verifiable multiauthority system was proposed that allows nonvoting even if a voter has registered. Also, it supports tallying by providing privacy to voters. To diffuse the power of a single authority, distributed blind signatures (Chaum, 1984) are used. The architecture described consists of seven phases with voters, administrators, scrutineers, and counter. The votes are encrypted, then signed by the administrator using a blind threshold signature. The voters then create the encrypted votes and send them to the counter using an untraceable email system. The final votes are sent to the scrutineer, who then confirms and sends votes back with their secret key. Finally, the decrypted votes are published. Voter privacy is secured in this scheme. The time complexity of the system is linear with respect to the number of voters per administrator.

In Li et al. (2009), the authors present a multiauthority voting system based on blind signatures. Voters cast their vote and the vote is signed by multiple authorities to blind it. The authorities are responsible for an honest and fair election by conforming with government agencies. The protocol has four phases, and four trusted authorities have three pairs of keys. Each voter has two pairs of keys. The main contribution of the paper is that coercion resistance and identification of a voter are not fully dependent on the trusted authorities.

Authors in Porkodi et al. (2011) present a scheme with a bulletin board where the voters post their encrypted votes. The homomorphic properties of encryption allow an anonymous tally of votes from the bulletin board. A threshold scheme is used for key sharing among the administrators. A commitment needs to be posted by each administrator to his/her private key and prove that a voter has a valid, encrypted vote. In the final stage, the tally is decrypted and the results are published. For a smaller key size, elliptic curve cryptosystems (Miller, 1985; Hankerson et al., 2006) are used. Each vote comes with a proof of knowledge, and administrators have to prove the validity of their commitment.

A multiauthority, receipt-free system is proposed by Philip et al. (Adewole et al., 2011) where n authorities are available and a trusted center (TC) is present. The TC is responsible for providing username and password to eligible voters after validating their identities and registering their votes. Each voter has to encrypt their vote, sign it, and craft the proof of correctness. Once that process is over, authorities distribute the shared private key and get the final tally. To provide receipt-less voting, the votes are encrypted again, signed, and have a proof of validity. The computational cost of the protocol is very high.

Cobra by Essex et al. (2012) used the idea of the proof of concept that simultaneously provides ballot authorization. To achieve lack of coercibility, each elector can use a fake credential and cast fake votes, which are finally separated out at the end without revealing any other submission (Schläpfer et al., 2011). Bloom Filter (Bloom, 1970) is used to prevent privacy of the voters. The concurrent ballot authorization is computationally high. Authorities are assumed to be honest and trustworthy. The system also faces scalability issues.

In another work (Nguyen and Dang, 2013), the authors propose a blind signature-based voting scheme using dynamic ballots. The registration of a voter is done using blind signatures and a group of authorities. A dynamic ballot changes for each voter, so it ensures lack of coercibility. Each voter receives a random permutation of the candidates, which is done by the ballot center. Plain-text-equivalence (PET) (Juels et al., 2010) is used for tallying purpose, where invalid and repeated ballots are removed. This protocol satisfies every need for a secured voting system. However, with five different authorities, two sets of bulletin boards, two forms of encryption, and a PET there come scalability issues, and this results in high computation cost.

Aziz (2019) shows an e-voting work with blind signatures that has multiple authorities and is coercion resistant. Freedom from coercion is employed by using a scheme similar to that in Essex et al. (2012). Once the registration process is over, a token authority is responsible for providing tokens to the voters for their anonymous request. It is further used to create a ballot. This ensures receipt-less voting as ballot-making is not in the hands of the voters. The ballots are further signed by a registrar using blind signatures, and these ballots are cast through mixnets. A set of trusted parties is present for decrypting the votes and calculating the final tally. The vote and the proof of validity are delinked by shuffling the ballots, thus preventing the voter from knowing if the vote was tallied or not. This provides resistance to coercion.

Tables 9.1 and 9.2 comprise a comparison of the properties of the various works discussed in the preceding section. The works are compared against the security properties and performance properties required for a secured e-voting system.

2.4 Issues in e-voting

Even though e-voting systems provably increase voter turnout, helping to make the election fair and free from fraud, many gaps remain in the currently used e-voting systems that need to be addressed. Most of the

Table 9.1 Security properties comparison of e-voting systems.

Approaches	Eligibility	Privacy	Voter anonymity	Lack of reusability	Verifiability	Fairness	Lack of coercibility	Receipt-less
Clarke and Martens (2016)	Yes	Yes	Yes	Yes	Yes	Yes	—	—
D.L.Chaum (1981)	No	Yes	Yes	No	Yes	Yes	No	No
Cohen and Fischer (1985)	No	No	Yes	No	No	Yes	No	No
J.D.Cohen (1986)	No	Yes	Yes	No	No	Yes	No	No
Benaloh and Tunistra (1994)	—	Yes	—	—	Yes	Yes	—	Yes
Cramer et al. (1996)	—	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cramer et al. (1997)	—	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Juang et al. (2002)	—	Yes	Yes	—	Yes	Yes	—	—
Li et al. (2009)	Yes	Yes	Yes	—	—	—	Yes	Yes
Porkodi et al. (2011)	Yes	Yes	Yes	Yes	Yes	—	—	—
Adewole et al. (2011)	Yes	Yes	Yes	—	Yes	Yes	Yes	Yes
Essex et al. (2012)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nguyen and Dang (2013)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Aziz (2019)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Note: "—", not available; *No*, property not supported; *Yes*, property supported.

Table 9.2 Performance properties comparison of e-voting systems.

Approaches	Correctness	Robustness	Efficiency	Mobility
Clarke and Martens (2016)	Yes	Yes	Yes	Yes
Chaum (1981)	Yes	No	Yes	No
Cohen and Fischer (1985)	Yes	Yes	Yes	No
Cohen (1986)	Yes	Yes	Yes	No
Benaloh and Tunistra (1994)	Yes	No	Yes	No
Cramer et al. (1996)	Yes	Yes	Yes	No
Cramer et al. (1997)	Yes	Yes	Yes	No
Juang et al. (2002)	Yes	Yes	Yes	No
Li et al. (2009)	Yes	Yes	Yes	No
Porkodi et al. (2011)	Yes	Yes	Yes	No
Adewole et al. (2011)	Yes	Yes	Yes	No
Essex et al. (2012)	Yes	Yes	Yes	No
Nguyen and Dang (2013)	Yes	Yes	Yes	No
Aziz (2019)	Yes	Yes	Yes	Yes

Note: *No*, property not supported; *Yes*, property supported.

current e-voting systems are centralized in nature, with just one point of trust. Such systems are prone to distributed denial of service (DDoS) attacks, resulting in the stoppage of the server that leads to the failure of the entire system. This leads to authentication issues followed by privacy problems (Bokslag & de Vries). In a majority of the systems, the databases are completely controlled by a single authority, which makes the system vulnerable to biasing, manipulation, and can lead to election fraud.

There have been studies that show online ballots being subjected to vote manipulation and impersonation of votes. When votes are electronically transmitted, it is difficult to detect if a vote is genuine or manipulated (Susskind, 2017). Another major issue is that the voting process, particularly the tallying phase, is not transparent, which entails the trust issue in voters' minds (Bokslag & de Vries; Oo and Aung, 2014). Also, there is no way to recount the ballots in case of any discrepancies. The aforementioned problems instigated researchers to think of alternate methods to secure the voting system.

3. Distributed e-voting

A distributed trust mechanism for the voting system can solve the trust-worthiness issues. The distributed trust model can be established using blockchain to provide transparency and immutability. In 1991, Haber and Stornetta (1991) had introduced the idea of time-stamping digital documents to make them tamper-proof. Something similar, based on the concept of blockchain, was developed in 2008 by Satoshi Nakamoto (Nakamoto). The first application of the concept was seen in Bitcoin, making a mark in the field of cryptocurrency (Homepage). Blockchain can be used in any field that involves any sort of transaction. Blockchains are tamper-resistant distributed ledgers (tamper evident) that are implemented without a central authority. The trust is distributed as each transaction has a joint source and needs to be validated by nodes. The system relies on the concept of asymmetric key cryptography (Nakamoto). A blockchain can be described in simple words as a chain of timestamped blocks that are linked by cryptographic hashes making them immutable and tamper evident. Each block holds the cryptographic hash of the previous block, and once a new block is added, modification of the older block becomes difficult, giving it the property of tamper resistance (Table 9.2).

With the properties that blockchain holds, it can provide the following to the e-voting system:

- Transparency, integrity, verifiability, and confidentiality are possible (Hjalmarsson et al., 2018; Hanifatunnisa and Rahardjo, 2017), as a voter or any independent individual can observe the results of the election without revealing the voter's choice nor identity that are on the blockchain. Blockchains are immutable so establish integrity in the system.
- Consistency is achieved, as the system is distributed in nature, so if any node is disrupted in service, it receives a copy of every transaction once it comes back online (Hjalmarsson et al., 2018; Pathak et al., 2018)
- Availability (Hjalmarsson et al., 2018; Pathak et al., 2018; Akbari et al., 2017) is provided as each node runs separately, so there is no single point of failure.
- Voter confidence is possible since by incorporating blockchain in the e-voting system the process will become noted and results can be published instantly. Moreover with its open-source and transparent nature, it will increase voter confidence and finally increase voter turnaround (Kshetri and Voas, 2018).

Similar to a general e-voting structure, there are various phases in a Blockchain-based e-voting systems. While the actors remain same, the working is a bit different, as depicted in Fig. 9.2. Each voter casts their vote after successfully completing the voter verification phase in a block. If a group of actors mine and verify, a consensus is formed, and the block is

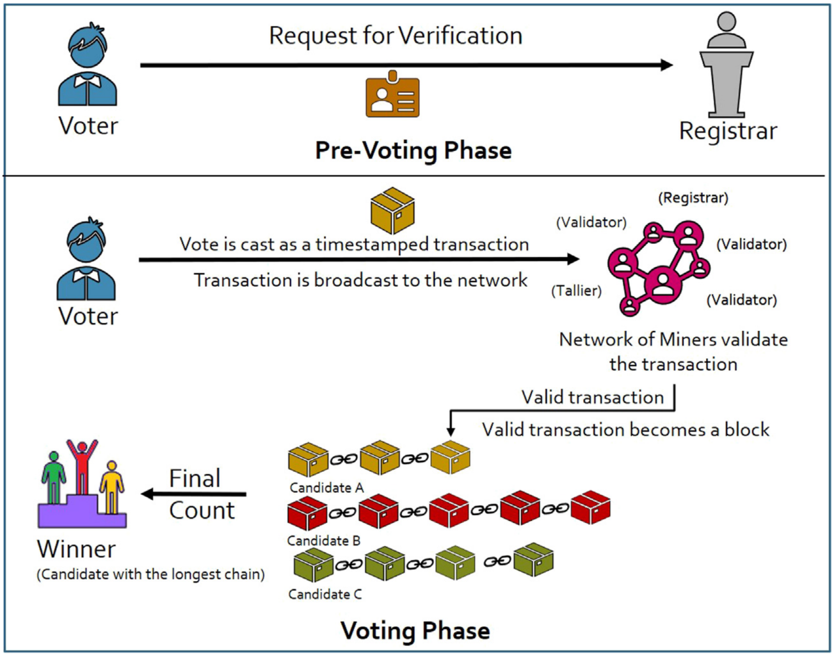

Figure 9.2 Typical phases in a blockchain-based e-voting protocol (Ayed, 2017).

added to the blockchain. Here, each candidate is dedicated to a blockchain, and the candidate with the longest chain length is declared as the winner by the tallier (Ayed, 2017).

3.1 Significant contributions in blockchain-based e-voting

Many researchers have pointed out that blockchain can be incorporated into e-voting to gain accountability and voter's trust in the process (Riemann and Grumbach, 2017; Schiedermeier et al., 2019; Dricot and Pereira, 2018). Riemann and Grumbach (2017) state that building a smart contract can provide an efficient voting protocol based on blockchain by using third-party features, so coercion can be prevented (Sudharsan et al., 2019; Teja et al., 2019; Sadia et al., 2019; Shahzad and Crowcroft, 2019). Many contributions claim that blockchain can produce a solution with respect to security and transparency in a voting system (Zhang et al., 2019; Heiberg et al., 2018; Srivastava et al., 2018b; Bosri et al., 2019).

In 2015, The Blockchain Technologies Corp. proposed a blockchain called VoteUnit (Ruoti et al., 2019) that functioned like BitCoin, except for the fact that it did not require a fee for any transaction. When a voter votes for a candidate, a payment is made to the selected candidate's wallet. The whole system was an offline procedure to safeguard the votes from getting manipulated. The system was able to provide an immutable trail of transactions for the voters and thus establish voter verifiability. But it was dependent on a TTP for voter registration.

In 2017, McCorry et al. (2017) were the first to design a decentralized boardroom voting protocol without using any TTP that supported self-tallying of votes. They used a smart contract that was used on the Open Vote Network (OVN) and ran it on Ethereum. Instead of using blockchain as just a bulletin board, they used it for its power of consensus computing. The system was built with Byzantine fault tolerance consensus. The issues with their systems were that only 50 voters could participate in voting and that there needed to be some trust in the central administration for voter registration and authentication.

Another blockchain-based voting system is FollowMyVote (Anonymous). It is a completely online process that uses Elliptic Curve Cryptography in the creation of votes and uses the blockchain technology as a ballot box. Here a voter has to download the FollowMyVote application on their desired devices and register by submitting his/her picture and government ID along with the Identity Key Pair. The voter then anonymously registers the Voter Key Pair. This ensures the voter's anonymity. Once the verifier approves a voter, the registrar is then responsible for allotting the correct ballot type to the voter. After voting, the ballots are submitted into a blockchain that ensures that the vote is immutable, transparent, and secured. A voter signs the ballot using the Voter Key Pair. The votes are thus anonymous and publicly available and allow any participant of the election to verify and tally the votes. The FollowMyVote application also permits a voter to change his/her vote up to a certain time as decided by the election officials. The system was efficient but not completely secured, as it is dependent on a TTP for voter registration and authentication, which can be compromised.

Lee et al. (2016) proposed the introduction of a TTP along with an authentication organization (AO) that generates the voter's list. This organization determines if a voter has the right to vote and verifies the voter's identity. They stated that an AO can link a vote back to its voter. Also, being the sole institution, it can manipulate the voter's list. In this paper the authors tried to prove that trusting a single organization can result in forgery and malpractice. In their proposed protocol, two different organizations produce their own list, so tallying both can prevent malpractices. Zhao and

Chan used a lottery protocol (Andrychowicz et al., 2014) along with zero-knowledge proof, which is used to distribute random numbers that hide the ballot (Zhao and Chan, 2016). Even though this was one of the first attempts, there were many limitations in this approach. The authors did not speak about voter verification nor authentication. Also it worked only for two candidates, and voter identification was also ignored. Proof of Vote was used by Li et al. (2017) to ensure privacy, accuracy, and reliability of the voting system.

Seifelnasr et al. (2020) tried to improve the OVN and introduced an off-chain administrator. There is no need to trust a single authority, and bulk computations can be performed by the off-chain administrator. Every voting requirement, namely, vote privacy, voter anonymity, etc., was fulfilled by the proposed protocol. Lack of reusability was not ensured in the system. Wang et al. (2018) used homomorphic encryption and ring signature for large-scale voting. They used smart contracts on Ethereum. The system provided anonymity but is neither robust nor resistant to coercion.

Kshetri and Voas (2018) addressed the challenges of fraudulent votes. They were the first to pay the voters for voting. A voter is allowed to vote only once, but is allowed to change his/her vote within the designated time of the voting process. The problem of fraudulent voting is tackled by the blockchain, as every transaction is visible on the ledger so can be detected by the consensus network. To manipulate an already cast vote, one has to hack all the older blocks, which is computationally difficult.

Tarasov andTewari (2017) proposed a payment-based system where the candidate receives payments as votes from a voter using the concept of ZCash (Bowe et al., 2017). They assumed that voter identities can be verified efficiently. The disadvantage of such a system is that a fraudulent voter might not make a payment to the candidate to hold on to their money. Along with this, the system needs a trusted authority that operates in a centralized manner.

Hardwick et al. (2018) used Ethereum blockchain for implementation of their proposed protocol. There was a trusted central authority (CA) that maintained the voter's list. The authors made a consideration that if multiple CAs could be incorporated, with each CA holding only partial voter information, impersonation attack can be handled. They allowed voters to change their votes until the end of the election. Every voting requirement was fulfilled in the proposed system.

Bistarelli et al. (2017) used the Bitcoin system to implement a decentralized voting system. They claim to have created a system that directly allows a voter to cast his vote in the blockchain. Here, a voter has a wallet containing a set of keys. These keys are used for signing transactions. To prevent the voter's public key from revealing a voter's identity, the authors propose the use of anonymous Kerberos, which provides an authentication mechanism without revealing the client's identity. The authors ensured that all the source addresses of the confirmed voters are remembered, thus ensuring that one vote is counted per authorized voter. When asset coins are used, the token can be marked with the attached metadata that is signed by the election commission. This converts a Bitcoin into a vote, so single votes can be counted. To prevent voters from sending asset coins to an unauthorized voter, the authors suggest that one can use a permissioned blockchain where transactions among voters are not permitted.

Ayed (2017) proposed an e-voting system incorporating blockchain. In the proposal, it is assumed that only registered voters participate in voting. A voter's Social Security number is used for authentication and to check if a voter is a valid. Once a voter is authorized, they are allowed to cast a vote. Once the vote is cast from the provided user interface in the voter's device, the system generates a unique input containing the voter's ID number, full name, and the hash of the previous vote. This input is then encrypted using SHA-256 and is stored in the block header of each vote that is cast. This ensures that voter information cannot be obtained from the votes. For each vote, a block gets created. Every time a candidate is voted for, a new block gets added to their corresponding blockchain. The authors have dealt with the problem of concusses that occurs when multiple voters vote at the same time. As a solution to this problem, The longest chain rule (followed by Bitcoin) is used. Each candidate has a different chain. The votes cast in the blockchain are public and verifiable. This protocol fulfills every requirement for secured voting.

Na and Park (2018) proposed a chat-based voting system. The system was implemented using Hyperledger Fabric. It used the idea of taking group decisions for voting purpose. It assured anonymity and fulfilled other voting properties. Bartolucci et al. (2018) proposed SHARVOT protocol that used circle shuffle, which requires a CA to control the whole process, thus bringing a single point of failure. This protocol prevented linking a voter's identity with their ballots. However, their system is vulnerable to multiple vote casting and only a limited number of participants were allowed.

Votereum (Thuy et al., 2019) was built on the Ethereum platform, where the authors tried to reduce trust on any CA. The proposed system provided privacy, robustness, and verifiability to the voters. The system however fails to be receipt-free and resistant to coercion. Also, the application server used for authorization is vulnerable to DDoS attack. In 2018, Hjalamrsson et al. (2018) proposed a scheme using private Ethereum blockchain and used smart contracts for tallying the result. They ensured transparency by allowing the voters to access the blockchain during voting.

Yavuz et al. (2018) developed a voting app on the android platform using Ethereum, but it failed to provide robustness and resistance to coercion. Quantum blockchain (Kiktenko et al., 2018) was used by Sun et al. (2019), which allows a classical channel to not be fully authenticated as the quantum channel is authenticated. They used quantum binding to achieve anonymity and fairness. Another group of authors (Srivastava et al., 2018a,b) used phantom protocol (Sompolinsky & Zohar) that is used to handle large transactions to attain the property of scalability. They also proved that their protocol was a fast one and took little time to produce results.

Lai and Wu (2018) used Ethereum blockchain to provide transparency to voters, who were able to view the results. Liu and Wang (2017) proposed a voting scheme without using any TTP. They use a combination of blind signature and blockchain in their protocol. Blind signature preserves a voter's choice and blockchain provides transparency to the election. A ballot is considered valid if it has the correct message format, has all the signatures, is cast on time, and has not been counted. The authors did not mention any tallying technique. After the tally process is over, the organizer publishes a list of all the valid ballots and the result. This information can be verified by all participants who have the permission to view the blockchain. There were some security issues in this protocol. The IP address of the voter can be obtained, which can expose the connection between the voter and ballot. Also, since the ballots can be viewed during the election process, it can affect the outcomes of the election. They fulfill every secured voting property except robustness, coercion resistance, and fairness.

Shahzad and Crowcroft (2019) proposed a system where the transactions are sealed with another hash function to provide integrity. Chaieb et al. (2019) proposed a system of identity-based encryption techniques ensuring all security features. Gao et al. (2019) designed a small-scale code-based voting protocol using public key cryptography. Luo et al. (2018) proposed a system using ring-based consensus algorithm, ensuring all security features.

On March 7, 2018, Sierra Leone became the first country in the world to conduct a blockchain-based election using the Agora platform (Chohan). Blockchain ensured that the election was transparent. Since a permissioned blockchain was used, votes were visible to all users, but validation was conducted only by authorized personnel. The country also saw a decline in electoral violence, so it can be used as a case study to prove the accountability of blockchain in an electoral process.

Tables 9.3 and 9.4 depict a comparative study of the aforementioned research works with respect to security properties and performance properties.

3.2 Some issues in blockchain-based e-voting

The advantages of blockchain-based e-voting lie in the protocol and how it has been implemented. The main issue arises when there is a possibility of violation of the election rules. These need to be very carefully specified in the smart contracts, so there is a dependency on an administrator or a TTP to incorporate the same.

Another issue arises when a public blockchain is used in a voting protocol. Anyone can access and see the transactions present there, thus creating a biased election. As a solution, a private blockchain can be used, which in turn questions transparency. Furthermore, scalability issues in blockchain are yet to be explored. Along with that the encryption algorithms that are used in such various protocols need to be robust.

Also, an election is an event where a lot of people are involved. They vote around the same time. This requires a system with high throughput. Existing blockchain networks, however, lack in this aspect, with Bitcoin having a throughput as low as 7 transactions per second (TPS) and Hyperledger Fabric having about 500 TPS (Huang et al., 2021).

Since blockchain addresses are generated randomly, a voter can vote anonymously, but it has been shown (Feng et al., 2019) that transactions can be traced back to revealing user identities and that users' pseudo-identities can be linked to their IP addresses (Biryukov et al., 2014), thus questioning the users' privacy and security.

Finally, an eligibility check and authentication process for a voter is mainly dependent on the TTP or a trusted authority, which still remains as a challenge for most of the blockchain-based voting protocols.

Table 9.3 Security properties comparison of blockchain-based e-voting systems.

Approaches	Eligibility	Privacy	Voter anonymity	Lack of reusability	Verifiability	Fairness	Lack of coercibility	Receipt-less
McCorry et al. (2017)	No	Yes	Yes	Yes	No	Yes	No	Yes
Kshetri and Voas (2018)	Yes	Yes	Yes	Yes	No	—	—	—
Hardwick et al. (2018)	Yes	Yes	Yes	No	—	—	—	—
Bistarelli et al. (2017)	No	Yes	Yes	No	—	—	No	—
Na and Park (2018)	Yes	Yes	Yes	Yes	—	—	—	—
Zhao and Chan (2016)	No	Yes	Yes	Yes	Yes	—	—	—
Ahmed Ben Ayed (2017)	Yes	Yes	Yes	Yes	No	Yes	No	Yes
FollowMyVote (Anonymous)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Seifelnasr et al. (2020)	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Bartolucci et al. (2018)	Yes	Yes	Yes	Yes	Yes	Yes	No	No
L.V.C Thuy et al. (2019)	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Hjalmarsson et al. (2018)	Yes	Yes	Yes	Yes	—	Yes	Yes	Yes
Teja et al. (2019)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Yavuz et al. (2018)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Zhang et al. (2019)	Yes	Yes	Yes	Yes	No	No	No	No
Tarasov and Tewari (2017)	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Sun et al. (2019)	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Srivastava et al. (2018a)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Lai et al. (2018)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wang et al. (2018)	Yes	Yes	Yes	Yes	Yes	Yes	No	No
Liu and Wang (2017)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Shahzad and Crowcroft (2019)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Murtaza et al. (2019)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Chaieb et al. (2019)	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Venkatpur et al. (2018)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Gao et al. (2019)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Luo et al. (2018)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Li et al. (2009)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Leonardos et al. (2020)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Kibin Lee et al. (2016)	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes

Note: "—", not available; *No*, property not supported; *Yes*, property supported.

Table 9.4 Performance properties comparison of blockchain-based e-voting systems.

Approaches	Correctness	Robustness	Efficiency	Mobility
McCorry et al. (2017)	Yes	Yes	Yes	Yes
Kshetri and Voas (2018)	Yes	Yes	Yes	Yes
Hardwick and Gioulis (2018)	Yes	Yes	Yes	Yes
Bistarelli et al. (2017)	Yes	Yes	Yes	Yes
Na and Park (2018)	Yes	Yes	Yes	Yes
Zhao and Chan (2016)	Yes	Yes	Yes	Yes
Ahmed Ben Ayed (2017)	Yes	Yes	Yes	Yes
FollowMyVote (Anonymous)	Yes	Yes	Yes	Yes
Seifelnasr et al. (2020)	Yes	Yes	Yes	Yes
Bartolucci et al. (2018)	Yes	Yes	Yes	Yes
L.V.C Thuy (2019)	Yes	Yes	Yes	Yes
Hjalmarsson et al. (2018)	Yes	Yes	Yes	Yes
Teja et al. (2019)	Yes	Yes	Yes	Yes
Yavuz et al. (2018)	Yes	No	Yes	Yes
Zhang et al. (2019)	Yes	Yes	Yes	Yes
Tarasov and Tewari (2017)	Yes	Yes	Yes	Yes
Sun et al. (2019)	Yes	Yes	Yes	Yes
Srivastava et al. (2018b)	Yes	Yes	Yes	Yes
Lai et al. (2018)	Yes	Yes	Yes	Yes
Wang et al. (2018)	Yes	No	Yes	Yes
Liu and Wang (2017)	Yes	Yes	Yes	Yes
Shahzad and Crowcroft (2019)	Yes	Yes	Yes	Yes
Murtaza et al. (2019)	Yes	Yes	Yes	Yes
Chaieb et al. (2019)	Yes	Yes	Yes	Yes
Venkatpur et al. (2018)	Yes	Yes	Yes	Yes
Gao et al. (2019)	Yes	Yes	Yes	Yes
Luo et al. (2018)	Yes	Yes	Yes	Yes
Li et al. (2009)	Yes	Yes	Yes	Yes
Leonardos et al. (2020)	Yes	Yes	Yes	Yes
Kibin Lee et al. (2016)	Yes	Yes	Yes	Yes

Note: *No*, property not supported; *Yes*, property supported.

Therefore, it can be observed that solutions toward scalability, throughput, authentication, voter privacy, vote and election integrity, and protocol security need to be inspected with respect to blockchain-based e-voting protocols.

4. Conclusion

The concept of voting has been in play for many decades now. With the advancement in technology in the past few years, electronic voting systems have gained popularity. The integrity of elections is subject to debates regarding security issues faced by the systems. We looked into the various conventional voting system and studied their drawbacks. The 2000 US presidential election challenged the integrity of such a voting system. It gave rise to the concept of electronic voting. Internet voting became popular among citizens because it provided them the opportunity to vote from the comfort of their homes. We studied various researches proposing efficient and secured e-voting protocols. However, there were a few drawbacks like a single point of failure and biased voting. With the introduction of the blockchain concept in 2008, researchers found a solution to the problems that were being faced with existing protocols.

Many researchers have been working toward securing such e-voting protocols using the concept of blockchain, a decentralized distributed architecture that provides data consistency, immutability, transparency, and security. In this chapter, we summarized various existing blockchain-based e-voting protocols that researchers have tried to implement to develop tamper-resistant, transparent, secured voting mechanisms and finally presented a comparative study of both e-voting protocols and blockchain-based e-voting protocols.

References

Adewole, A.P., Sodiya, A.S., Arowolo, O.A., 2011. A Receipt-free Multi-Authority E-Voting System.

Akbari, E., Wu, Q., Zhao, W., Arabnia, H.R., Yang, M.Q., 2017. From blockchain to internet-based voting. In: Proceedings of the 2017 International Conference on Computational Science and Computational Intelligence (CSCI), Las Vegas, NV, USA, pp. 218–221, pp. 14–16.

Andrychowicz, M., Dziembowski, S., Malinowski, D., Mazurek, L., 2014. Secure multi-party computations on bitcoin. In: 2014 IEEE Symposium on Security and Privacy, SP 2014, Berkeley, CA, USA, May, 2014, pp. 443–458, pp. 18–21.

https://followmyvote.com/.

Awad, M., Leiss, E.L., 2016. The evolution of voting: analysis of conventional and electronic voting systems. Int. J. Appl. Eng. Res. 11 (12), 7888–7896.

Ayed, A.B., 2017. A conceptual secure blockchain-based electronic voting system. Int. J. Netw. Secur. Appl. 9 (3), 01–09.

Aziz, A., 2019. Coercion-resistant e-voting scheme with blind signatures. In: 2019 Cybersecurity and Cyberforensics Conference (CCC). IEEE, pp. 143–151.

Baiardi F., Falleni A., Granchi R., Martinelli F., Petrocchi M., Vaccarelli A., Seas, a secure e-voting protocol: design and implementation. Comput. Secur. 24 (8), 642.

Bartolucci, S., Bernat, P., Joseph, D., 2018. SHARVOT: secret SHARe-based VOTing on the blockchain. In: Proceedings of the 1st international workshop on emerging trends in software engineering for blockchain, pp. 30–34.

Benaloh, J., Tuinstra, D., 1994. Receipt-free secret-ballot elections. In: Proceedings of the Twenty-Sixth Annual ACM Symposium on Theory of Computing.

Biryukov, A., Khovratovich, D., Pustogarov, I., 2014. Deanonymisation of clients in bitcoin p2p network. In: Proceedings of the 2014 ACM SIGSAC Conference on Computer and Communications Security, pp. 15–29.

Bistarelli, S., Mantilacci, M., Santancini, P., Santini, F., 2017. An end-to-end voting-system based on bitcoin. In: Proceedings of the Symposium on Applied Computing, pp. 1836–1841.

Bloom, B.H., 1970. Space/time trade-offs in hash coding with allowable errors. Commun. ACM 13 (7), 422–426.

Bokslag, W., de Vries, M., Evaluating E-Voting: Theory and Practice, arXiv preprint arXiv:1602.02509.

Bosri, R., Uzzal, A.R., Al Omar, A., Hasan, A.S.M.T., Bhuiyan, M.Z.A., 2019. Towards a privacy-preserving voting system through blockchain technologies. In: 2019 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC/PiCom/CBDCom/CyberSciTech), pp. 602–608.

Bowe, S., Hornby, T., Wilcox, N., 2017. Zcash Protocol Specification.

Brent, P., 2006. The australian ballot: not the secret ballot. Aust. J. Polit. Sci. 41 (1), 39–50.

Chaieb, M., Koscina, M., Yousfi, S., Lafourcade, P., Robbana, R.D., 2019. Distributed authorities using blind signature to effect robust security in e-voting. In: Prague, C.R. (Ed.), Proceedings of the 16th International Joint Conference on E-Business and Telecommunications, pp. 228–235, pp. 26–28.

Chaum, D.L., 1981. Untraceable electronic mail, return addresses, and digital pseudonyms. Commun. ACM 24 (2), 84–90.

Chaum, D., 1984. Blind Signature System, Advances in Cryptology. Springer, Boston, MA.

Chohan, U.W., Blockchain Enhancing Political Accountability? Sierra Leone 2018 Case, Sierra Leone.

Clarke, D., Martens, T., 2016. E-voting in estonia, Real-World Electronic Voting: Design, Analysis and Deployment, pp. 129–141.

Cohen, J.D., Fischer, M.J., 1985. A Robust and Verifiable Cryptographically Secure Election Scheme. Yale University. Department of Computer Science.

Cohen, J.D., 1986. Improving Privacy in Cryptographic Elections. Yale University, Department of Computer Science, New Haven, CT.

Cramer, R., Franklin, M., Schoenmakers, B., Yung, M., 1996. Multi-authority secret-ballot elections with linear work. In: International Conference on the Theory and Applications of Cryptographic Techniques. Springer, pp. 72–83.

Cramer, R., Gennaro, R., Schoenmakers, B., 1997. A secure and optimally efficient multi-authority election scheme. Eur. Trans. Telecommun. 8 (5), 481–490.

Cranor, L.F., 2003. In search of the perfect voting technology: No easy answers. In: Secure Electronic Voting. Springer, pp. 17–30.

Dricot, L., Pereira, O.S. U.L.a., 2018. And Their Impact on Voting Systems. arXiv arXiv:180108064.

ElGamal, T., 1985. A public key cryptosystem and a signature scheme based on discrete logarithms. IEEE Trans. Info Theory 31 (4), 469–472.

Epstein, J., 2007. Electronic voting. Computer 40 (8), 92–95.

Essex, A., Clark, J., Hengartner, U., 2012. Cobra: Toward concurrent ballot authorization for internet voting. EVT/WOTE 12.

Feng, Q., He, D., Zeadally, S., Khan, M.K., Kumar, N., 2019. A survey on privacy protection in blockchain system. J. Netw. Comput. Appl. 126, 45—58.

Fredman, L.E., 1967. The introduction of the Australian ballot in the United States. Aust. J. Polit. Hist. 13 (2), 204—220.

Gao, S., Zheng, D., Guo, R., Jing, C., Hu, C., 2019. An anti-quantum e-voting protocol in blockchain with audit function. IEEE Access 7 (16), 15304—11531.

Greene, K.K., Byrne, M.D., Everett, S.P., 2006. A Comparison of Usability between Voting Methods. EVT.

Haber, S., Stornetta, W.S., 1991. How to time-stamp a digital document. J. Cryptol. 3, 99—111.

Hanifatunnisa, R., Rahardjo, B., 2017. Blockchain based e-voting recording system design. In: Proceedings of the 2017 11th International Conference on Telecommunication Systems Services and Applications (TSSA), Lombok, Indonesia, pp. 1—6, pp. 26—27.

Hankerson, D., Menezes, A.J., Vanstone, S., 2006. Guide to Elliptic Curve Cryptography. Springer, Science & Business Media.

Hardwick, F.S., Gioulis, A., Akram, R.N., Markantonakis, K.E.-V.w.B., 2018. An E-Voting Protocol with Decentralisation and Voter Privacy. arXiv arXiv:180510258.

Heiberg, S., Kubjas, I., Siim, J., Willemson, J., 2018. On Trade-Offs of Applying Blockchains for Electronic Voting Bulletin Boards, International Association for Cryptologic Research Available Online. URL. https://eprint.iacr.org/2018/685.pdf.

Hjalmarsson, F.P., Hreioarsson, G.K., Hamdaqa, M., Hjálmtýsson, G., 2018. Blockchain-based e-voting system. In: 11th IEEE International Conference on Cloud Computing, CLOUD 2018, San Francisco, CA, USA, July 2—7, 2018. IEEE Computer Society, pp. 983—986. https://doi.org/10.1109/CLOUD.2018.00151. URL.

Homepage, B., Available online: URL, https://bitcoin.org/.

Huang, J., He, D., Obaidat, M.S., Vijayakumar, P., Luo, M., Choo, K.-K.R., 2021. The application of the blockchain technology in voting systems: a review. ACM Comput. Surv. 54 (3), 1—28.

Joaquim, R., Zúquete, A., Ferreira, P., 2003. Revs—a robust electronic voting system. IADIS Int. J. WWW/Internet 1 (2), 47—63.

Jones, D.W., 2003. The evaluation of voting technology. In: Secure Electronic Voting. Springer, pp. 3—16.

Juang, W.-S., Lei, C.-L., Liaw, H.-T., 2002. A verifiable multi-authority secret election allowing abstention from voting. Comput. J. 45 (6), 672—682.

Juels, A., Catalano, D., Jakobsson, M., 2010. Coercion-resistant electronic elections. In: Towards Trustworthy Elections. Springer, Heidelberg, pp. 37—63. Berlin.

Kiktenko, E.O., Pozhar, N.O., Anufriev, M.N., Trushechkin, A.S., Yunusov, R.R., Kurochkin, Y.V., Lvovsky, A., Fedorov, A.K., 2018. Quantum-secured blockchain. Quant. Sci. Technol. 3 (3), 035004.

Kohno, T., et al., 2004. Analysis of an electronic voting system. In: on Security, I.S. (Ed.), Privacy, 2004, Proceedings. 2004. IEEE.

Kshetri, N., Voas, J.B.-E.E.-V., 2018. IEEE softw 35, 95—99.

Lai, W.-J., Hsieh, Y., Hsueh, C.-W., Wu, J.-L.D.A.D., 2018. Anonymous. In: Proceedings, I. (Ed.), Transparent E-Voting System, of the 2018 1st IEEE International Conference on Hot Information-Centric Networking (HotICN), Shenzhen, China, pp. 24—29, pp. 15—17.

Lambrinoudakis, C., Gritzalis, D., Tsoumas, V., Karyda, M., Ikonomopoulos, S., 2003. Secure electronic voting: the current landscape. In: Secure Electronic Voting. Springer, pp. 101—122.

Lauer, T.W., 2004. The risk of e-voting. Electron. J. eGovernment 2, 177—186.

Lee, K., James, J.I., Ejeta, T.G., Kim, H.J., 2016. Electronic voting service using blockchain. J. Digi. Foren. Secur. Law 11 (2), 8.
Leonardos, S., Reijsbergen, D., Piliouras, G., 2020. Weighted voting on the blockchain: improving consensus in proof of stake protocols. Int. J. Netw. Manag. 30 (5), e2093.
Li, C.-T., Hwang, M.-S., Lai, Y.-C., 2009. A verifiable electronic voting scheme over the internet. In: 2009 Sixth International Conference on Information Technology: New Generations. IEEE, pp. 449−454.
Li, J., Liu, Z., Chen, L., Chen, P., Wu, J., 2017. Blockchain-based security architecture for distributed cloud storage. In: Proceedings of the 2017 IEEE International Symposium on Parallel and Distributed Processing with Applications and 2017 IEEE International Conference on Ubiquitous Computing and Communications. ISPA/IUCC), Guangzhou, China, pp. 408−411, pp. 12−15.
Liu, Y., Wang, Q., 2017. An E-Voting Protocol Based on Blockchain, International Association for Cryptologic Research Available online. URL. https://eprint.iacr.org/2017/1043.pdf.
Luo, Y., Chen, Y., Chen, Q., Liang, Q.A., 2018. New election algorithm for dpos consensus mechanism in blockchain. In: Proceedings of the 2018 7th International Conference on Digital Home (ICDH), Guilin, China, 30 November-1, pp. 116−120.
Maaten, E., 2004. Towards Remote E-Voting: Estonian Case, Electronic Voting in Europe-Technology, Law, Politics and Society, Workshop of the ESF TED Programme Together with GI and OCG. Gesellschaft fur Informatik eV.
McCorry, P., Shahandashti, S.F., Hao, F.A., 2017. Smart contract for boardroom voting with maximum voter privacy. In: Cryptography, F., Security, D. (Eds.), Lecture Notes in Computer Science, Kiayias, A, Cham, Switzerland Volume 10322. -3-319-70971-0. Springer, pp. 357−375.
Miller, V.S., 1985. Use of elliptic curves in cryptography. In: Conference on the Theory and Application of Cryptographic Techniques. Springer, Heidelberg. Berlin.
Miller, W.R.H.C.M., 1995. Election fraud in late 19th century texas. Reg. Local Hist. 7, 111−128. URL. http://history.smsu.edu.
Murtaza, M.H., Alizai, Z.A., Iqbal, Z., 2019. Blockchain based anonymous voting system using zksnarks. In: Proceedings of the 2019 International Conference on Applied and Engineering Mathematics. ICAEM, Taxila, Pakistan, pp. 209−214, pp. 27−29.
Na, S., Park, Y.B., 2018. Web-based nominal group technique decision making tool using blockchain. In: Proceedings of the 2018 International Conference on Platform Technology and Service (PlatCon), Jeju, Korea, pp. 1−6, pp. 29−31.
Nakamoto, S.B.A., Peer-to-Peer Electronic Cash System, Available online. URL, https://bitcoin.org/bitcoin.pdf.
Nguyen, T.A.T., Dang, T.K., 2013. Enhanced security in internet voting protocol using blind signature and dynamic ballots. Electron. Commer. Res. 13 (3), 257−272.
Oo, H.N., Aung, A.M., 2014. A survey of different electronic voting systems. Int. J. Sci. Eng. Technol. Res. 3 (16), 3460−3464.
Pathak, A., Wasay, A., Singh, C., Bhavan, R., Umale, J., 2018. Design anddesign implementation of a secure and robust voting system based on blockchain. Int. J. Advert. 4, 869−875.
Porkodi, C., Arumuganathan, R., Vidya, K., 2011. Multi-authority electronic voting scheme based on elliptic curves. IJ Netw. Secur. 12 (2), 84−91.
Raikov, A., 2018. Accelerating technology for self-organising networked democracy. Futures 103, 17−26. https://doi.org/10.1016/j.futures.2018.03.015. URL.
Riemann, R., Grumbach, S., 2017. Distributed Protocols at the Rescue for Trustworthy Online Voting. arXiv arXiv:170504480.
Roth, S.K., 1998. Disenfranchised by design: voting systems and the election process. Inf. Des. J. 9 (1), 29−38.

Ruoti, S., Kaiser, B., Yerukhimovich, A., Clark, J., Cunningham, R., 2019. Blockchain technology: what is it good for? Commun. ACM 63 (1), 46–53.

Sadia, K., Masuduzzaman, M., Paul, R.K., Islam, A., 2019. Blockchain Based Secured E-Voting by Using the Assistance of Smart Contract. arXiv arXiv:191013635.

Saltman, R.G., et al., 2003. Public confidence and auditability in voting systems. In: Secure Electronic Voting. Springer, pp. 125–137.

Schiedermeier, M., Hasan, O., Mayer, T., Brunie, L., Kosch, H.A., 2019. Transparent Referendum Protocol with Immutable Proceedings and Verifiable Outcome for Trustless Networks. arXiv arXiv: 190906462.

Schläpfer, M., Haenni, R., Koenig, R., Spycher, O., 2011. Efficient vote authorization in coercion-resistant internet voting. In: International Conference on E-Voting and Identity. Springer, pp. 71–88.

Seifelnasr, M., Galal, H.S., Youssef, A.M., 2020. Scalable open-vote network on ethereum. In: International Conference on Financial Cryptography and Data Security. Springer, pp. 436–450.

Shahzad, B., Crowcroft, J., 2019. Trustworthy electronic voting using adjusted blockchain technology. IEEE Access 7, 24477–24488.

Sompolinsky, Y., Zohar, A. Phantom, IACR Cryptology ePrint Archive, Report 2018/104.

Srivastava, G., Dwivedi, A.D., Singh, R., 2018a. Phantom protocol as the new crypto-democracy. In: a. K., H. (Ed.), Computer Information Systems and Industrial Management; Saeed, Cham, Switzerland Volume 11127. -3-319-99953-1; Lecture Notes in Computer Science. Springer, pp. 499–509.

Srivastava, G., Dwivedi, D., Singh, A., 2018b. Crypto-democracy: decentralized voting scheme using blockchain technology. In: Proceedings of the 15th International Joint Conference on E-Business and Telecommunications, Porto, Portugal, pp. 674–679, pp. 26–28.

Sudharsan, B., Nidhish Krishna, M.P., Alagappan, M., et al., 2019. Secured electronic voting system using the concepts of blockchain. In: 2019 IEEE 10th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON). IEEE, pp. 0675–0681.

Sun, X., Wang, Q., Kulicki, P.A., 2019. Simple voting protocol on quantum blockchain. Int. J. Theor. Phys. 58, 275–281.

Susskind, J.D.D., 2017. Incentivizing blockchain voting technology for an improved election system. San. Diego Law Rev. 54, 785–821.

Tarasov, P., Tewari, H., 2017. Internet voting using zcash. IACR Cryptol. ePrint Arch. 2017, 585.

Teja, K., Shravani, M., Simha, C.Y., Kounte, M.R., 2019. Secured voting through blockchain technology. In: Proceedings of the 2019 3rd International Conference on Trends in Electronics and Informatics (ICOEI), Tirunelveli, India, pp. 1416–1419, pp. 23–25.

Thuy, L.V.-C., Cao-Minh, K., Dang-Le-Bao, C., Nguyen, T.A.V., 2019. An ethereum-based e-voting system. In: Proceedings of the 2019 IEEE-RIVF International Conference on Computing and Communication Technologies (RIVF), Danang, Vietnam, pp. 1–6, pp. 20–22.

Venkatapur, R.B., Prabhu, B., Navya, A., Roopini, R., Niranjan, S.A., 2018. Electronic voting machine based on blockchain technology and aadhar verification. Int. J. Innov. Eng. 3, 12–15.

Wang, B., Sun, J., He, Y., Pang, D., Lu, N., 2018. Large-scale election based on blockchain. Procedia Comput. Sci. 129, 234–237.

Yavuz, E., Koc, A.K., Cabuk, U.C., Dalkilic, G., 2018. Towards secure e-voting using ethereum blockchain. In: Proceedings of the 2018 6th International Symposium on Digital Forensic and Security (ISDFS), Antalya, Turkey, pp. 1–7, pp. 22–25.

Zhang, Q., Xu, B., Jing, H., Zheng, Z.Q.-C., 2019. An Ethereum Based E-Voting System. arXiv arXiv:190505041.

Zhao, Z., Chan, T.-H.H., 2016. How to vote privately using bitcoin. In: O, S., K, E., L, a.K. (Eds.), Information and Communications Security; Qing, Cham, Switzerland Volume 9543, 3-319-29813-9, Lecture Notes in Computer Science. Springer, pp. 82–96.